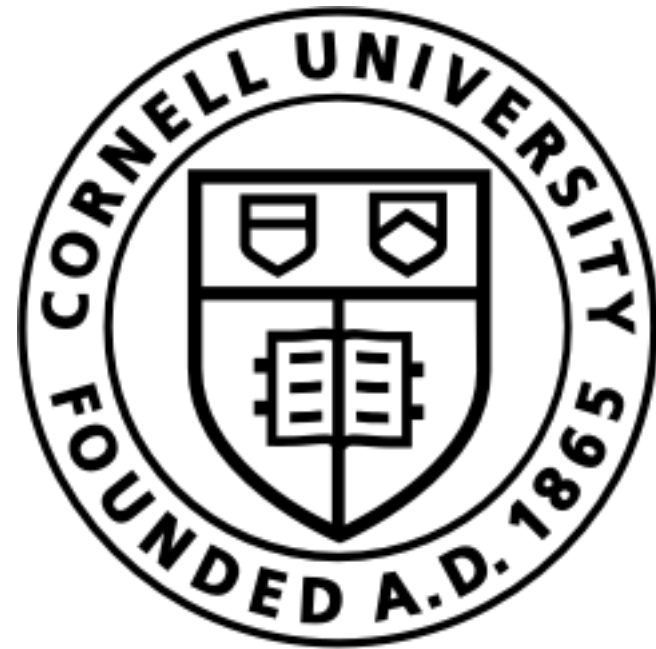




Lecture 3a: Analytical Analysis of Optimal Rational Choices and the Role of Resource Constraints

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Objectives. In this lecture, we will:

- ▶ **O1:** Introduce Constrained Resource Allocation Problems. Constrained resource allocation problems are found in a huge variety of disciplines, ranging from the sciences and engineering to hospitality.
- ▶ **O2:** The role of budget (and potentially) other constraints in resource allocation problems. Constraints shrink the space of possibilities to a specific feasible region.
- ▶ **O3:** Introduce the Lagrange multiplier method for solving constrained resource allocation problems: Maximize Utility subject to a budget constraint
- ▶ L3a and L3b long-form notes: <https://bit.ly/45EInfO>

Warning: Once you truly see constrained resource allocation problems, you can never unsee them.

Everything everywhere always becomes a constrained resource allocation problem.

Concept: Constrained Resource Allocation Problems

An agent has a set of n objects $X = \{x_i\}_{i=1}^n$, a utility function $U : X \rightarrow \mathbb{R}$, and a total of I units of resource to allocate, e.g., money, time, etc. An optimal rational agent maximizes its utility subject to its resource budget:

$$\text{maximize } \mathcal{O} = U(x_1, \dots, x_n) \quad (1)$$

$$\text{subject to } \sum_{i \in 1, \dots, n} c_i x_i = I \quad (2)$$

$$\text{and } x_i \geq 0 \quad i = 1, 2, \dots, n \quad (3)$$

The quantity $c_i \geq 0 \forall i$ denotes the cost of object i , and $x_i \geq 0$ represents the amount of object i purchased or consumed by the agent.

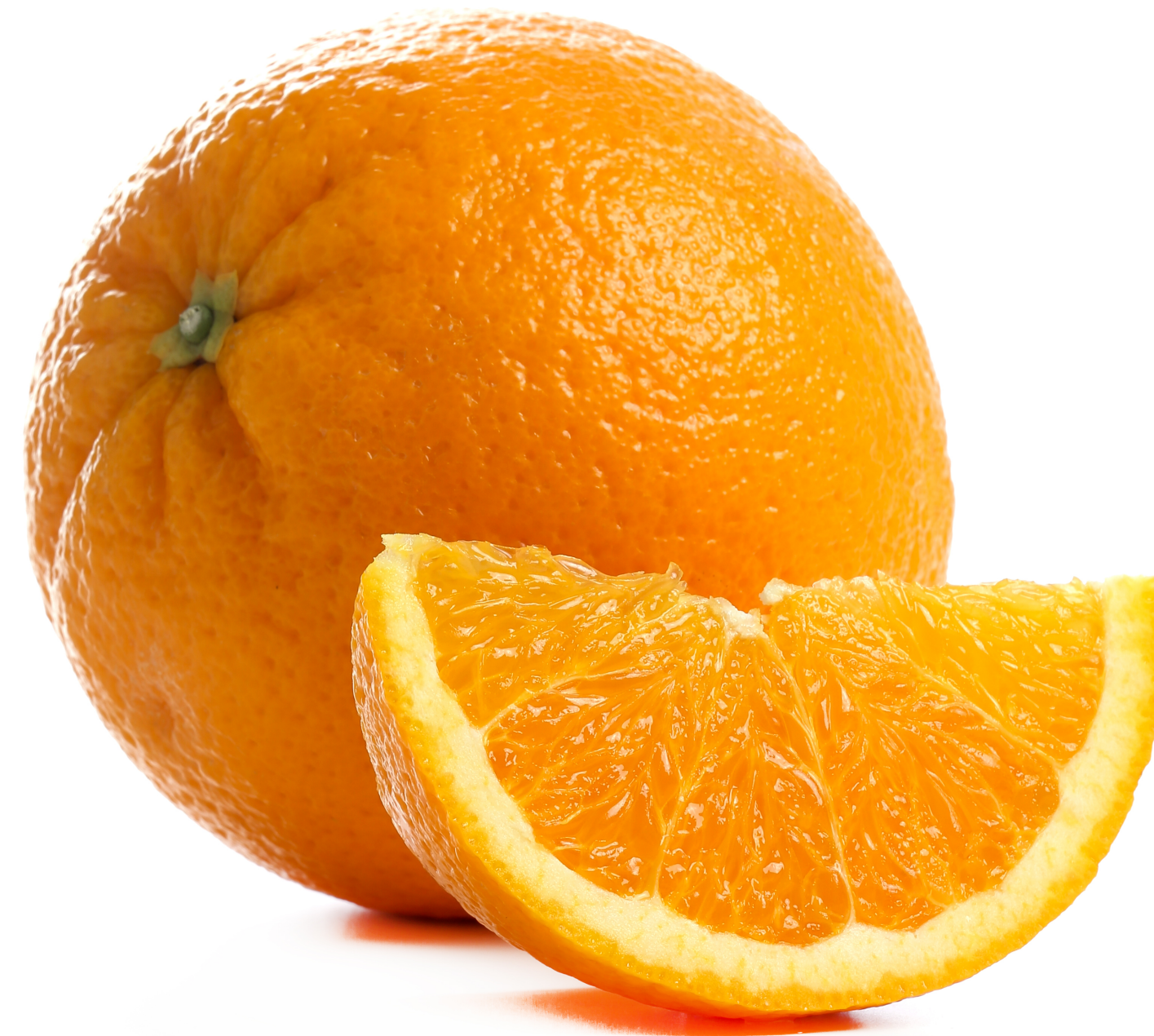
Concept: The Grocery Store Fruit Problem

Apples



x_1

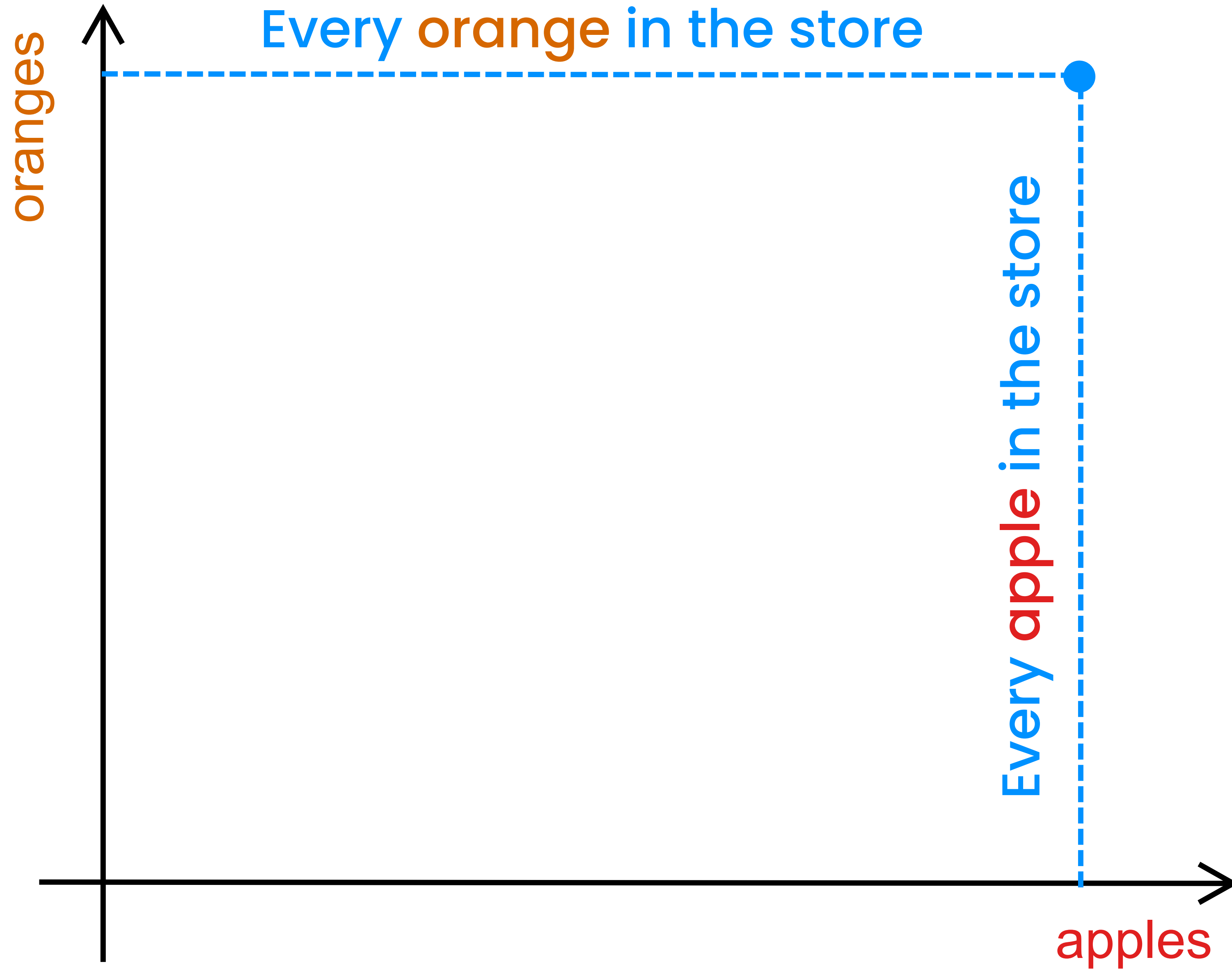
Oranges



- VS -

x_2

Rational Choice: Utility Analysis says More is Always Better

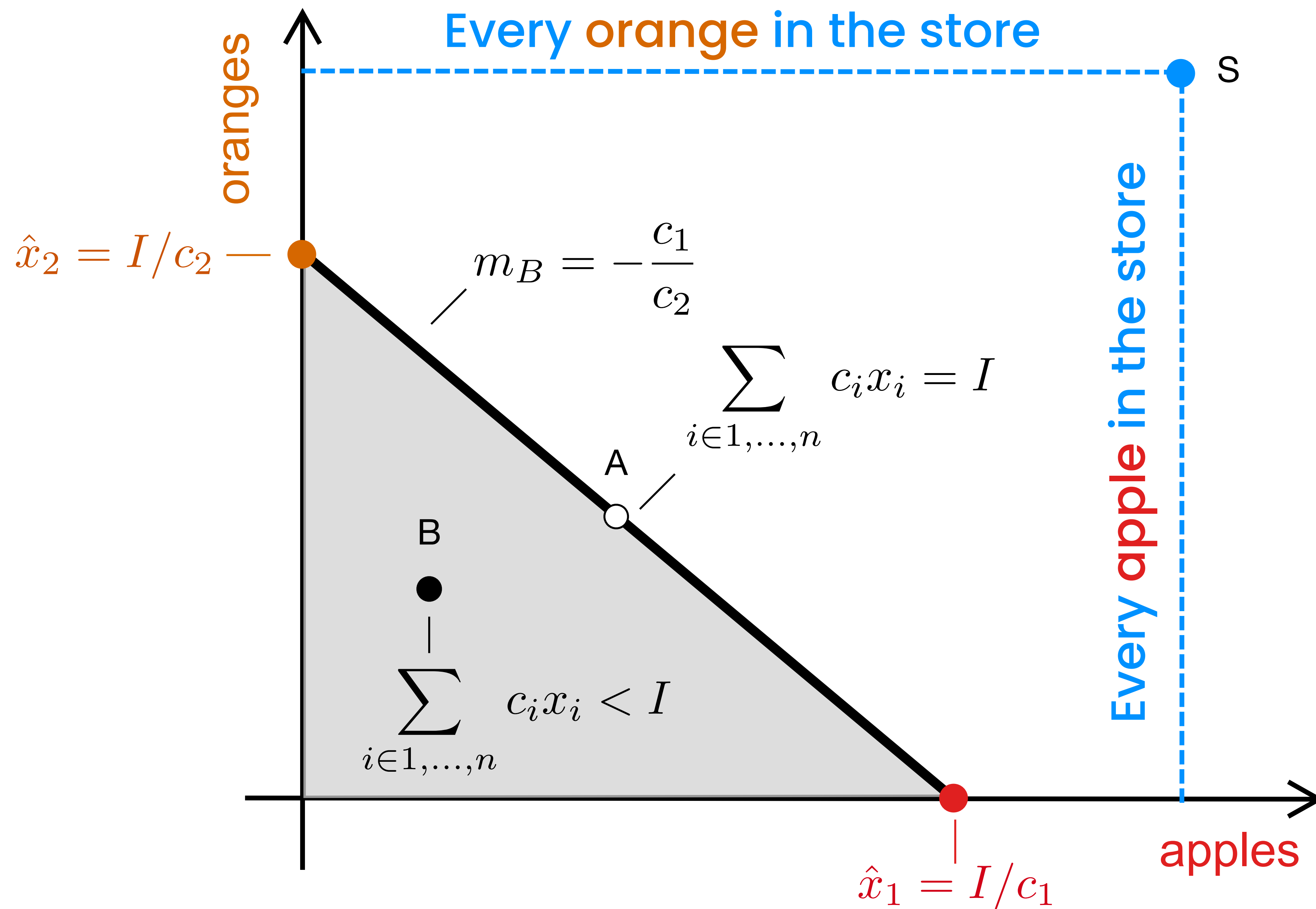


Concept: Resource Constraints

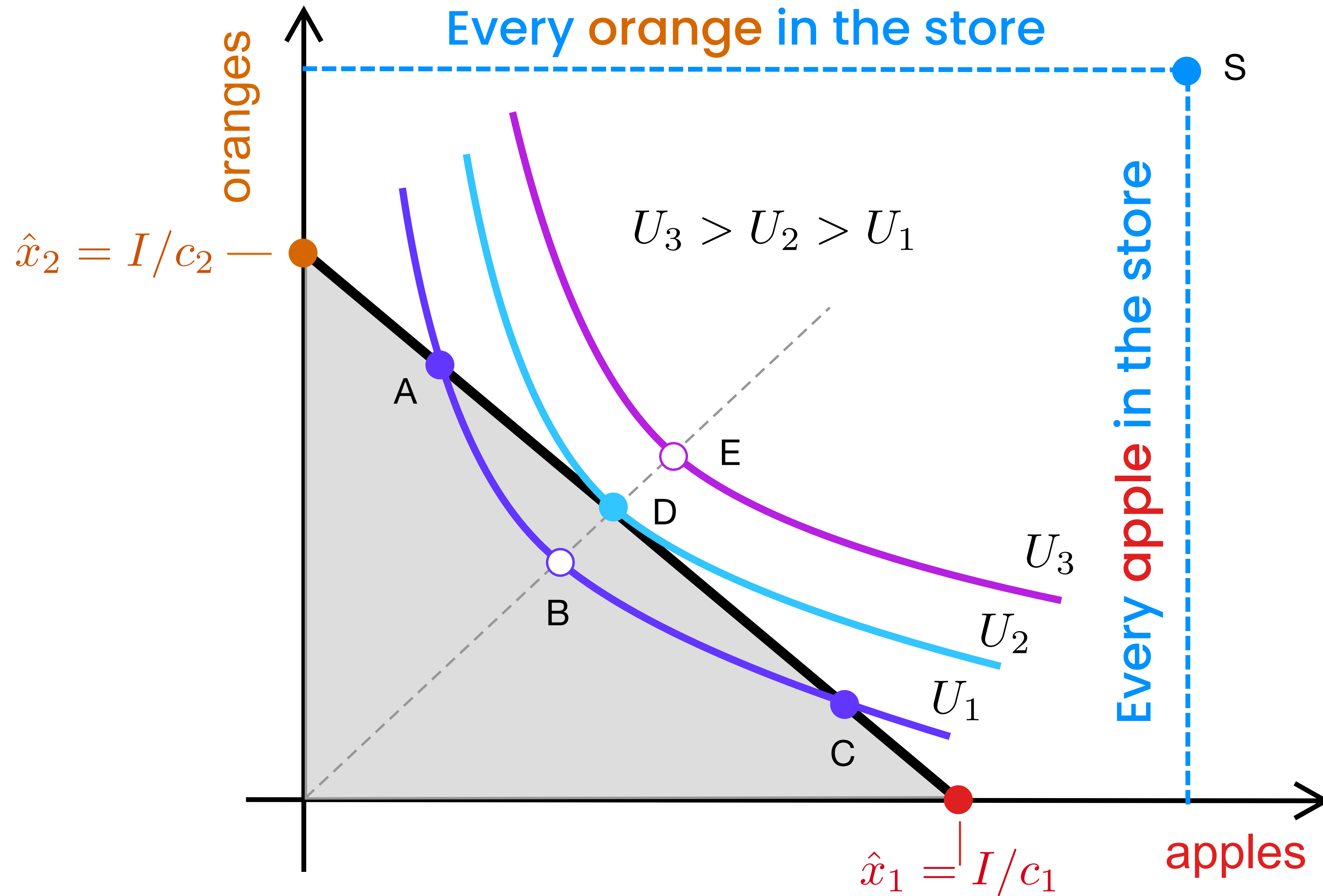
Let $I \geq 0$ denote a scarce resource, e.g., time, money, etc., that is allocated to consume, purchase, etc., the set of n objects $X = \{x_i\}_{i=1}^n$. The budget constraint for the set X is given by:

$$\sum_{i \in 1, \dots, n} c_i x_i = I \quad (4)$$

The terms $x_i \geq 0$ denote the quantity of object i consumed, purchased, etc, and $c_i \geq 0$ denotes the unit cost of object i .



Utility: Cobb-Douglas



O3: Introduce the Lagrange multiplier method for solving constrained resource allocation problems: **Maximize Utility subject to a budget constraint**

To find the optimum of a function $f(x)$ subject to the equality constraint $g(x)$, we form the Lagrangian $\mathcal{L}(x, \lambda)$:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda \cdot g(x) \quad (4)$$

where λ is the Lagrange multiplier for constraint $g(x)$. At a critical point (maximum or minimum), the partial derivatives of the Lagrangian with respect to x and λ vanish:

$$\frac{\partial \mathcal{L}}{\partial x} = f'(x) + \lambda \cdot g'(x) = 0 \quad (5)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = g(x) = 0 \quad (6)$$

This system of equations, which represents first-order optimality conditions, can be solved for the critical points and, thus, the maximum or minimum value of the objective function.

The constrained utility maximization problem has the Lagrangian:

$$\mathcal{L}(x, \lambda) = U(x_1, \dots, x_n) + \lambda \cdot \left(I - \sum_{i \in 1, \dots, n} c_i x_i \right)$$

which then gives the system of equations:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_i} &= \bar{U}_i - \lambda \cdot c_i = 0 & i = 1, 2, \dots, n \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= I - \sum_{i \in 1, \dots, n} c_i x_i = 0 \end{aligned}$$

where $\bar{U}_i$ denotes the marginal utility of good or service i . This $n + 1$ system of equations can be solved for the optimal values of x_i and λ .

Consider a utility maximization problem with the Cobb-Douglas utility function, $U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, subject to a budget constraint. The Lagrangian is:

$$\mathcal{L}(x, \lambda) = x_1^\alpha x_2^{1-\alpha} + \lambda \cdot (I - c_1 x_1 - c_2 x_2)$$

which gives the first-order optimality conditions:

$$\frac{\partial \mathcal{L}}{\partial x_1} = \alpha \cdot x_1^{\alpha-1} x_2^{1-\alpha} - \lambda \cdot c_1 = 0$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = (1 - \alpha) \cdot x_1^\alpha x_2^{-\alpha} - \lambda \cdot c_2 = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - c_1 x_1 - c_2 x_2 = 0$$

This system can be solved for the optimal consumption levels of x_1 and x_2 . However, it is non-linear. Thus, we often need to use numerical methods to find the optimal values of x_1 and x_2 .

Consider a utility maximization problem that uses a linear utility function $U(x_1, x_2) = \alpha_1 x_1 + \alpha_2 x_2$ subject to a budget constraint. The Lagrangian is:

$$\mathcal{L}(x, \lambda) = \alpha_1 x_1 + \alpha_2 x_2 + \lambda \cdot (I - c_1 x_1 - c_2 x_2)$$

which gives the first-order optimality conditions:

$$\frac{\partial \mathcal{L}}{\partial x_1} = \alpha_1 - \lambda \cdot c_1 = 0$$

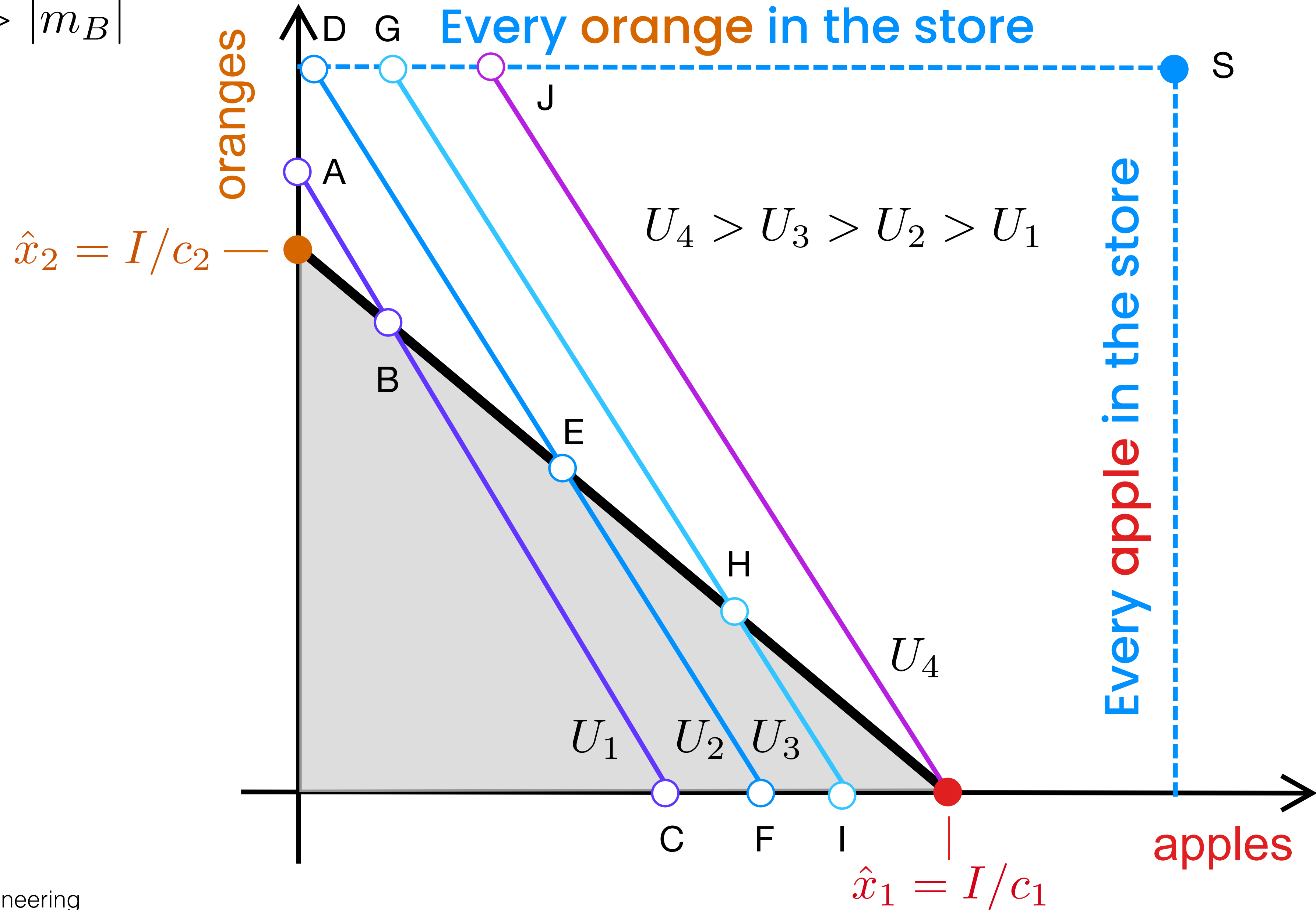
$$\frac{\partial \mathcal{L}}{\partial x_2} = \alpha_2 - \lambda \cdot c_2 = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - c_1 x_1 - c_2 x_2 = 0$$

This is a 3×3 system which is (unfortunately) underdetermined. Thus, the method of Lagrange multipliers suggests an infinite number of solutions (which is not true).

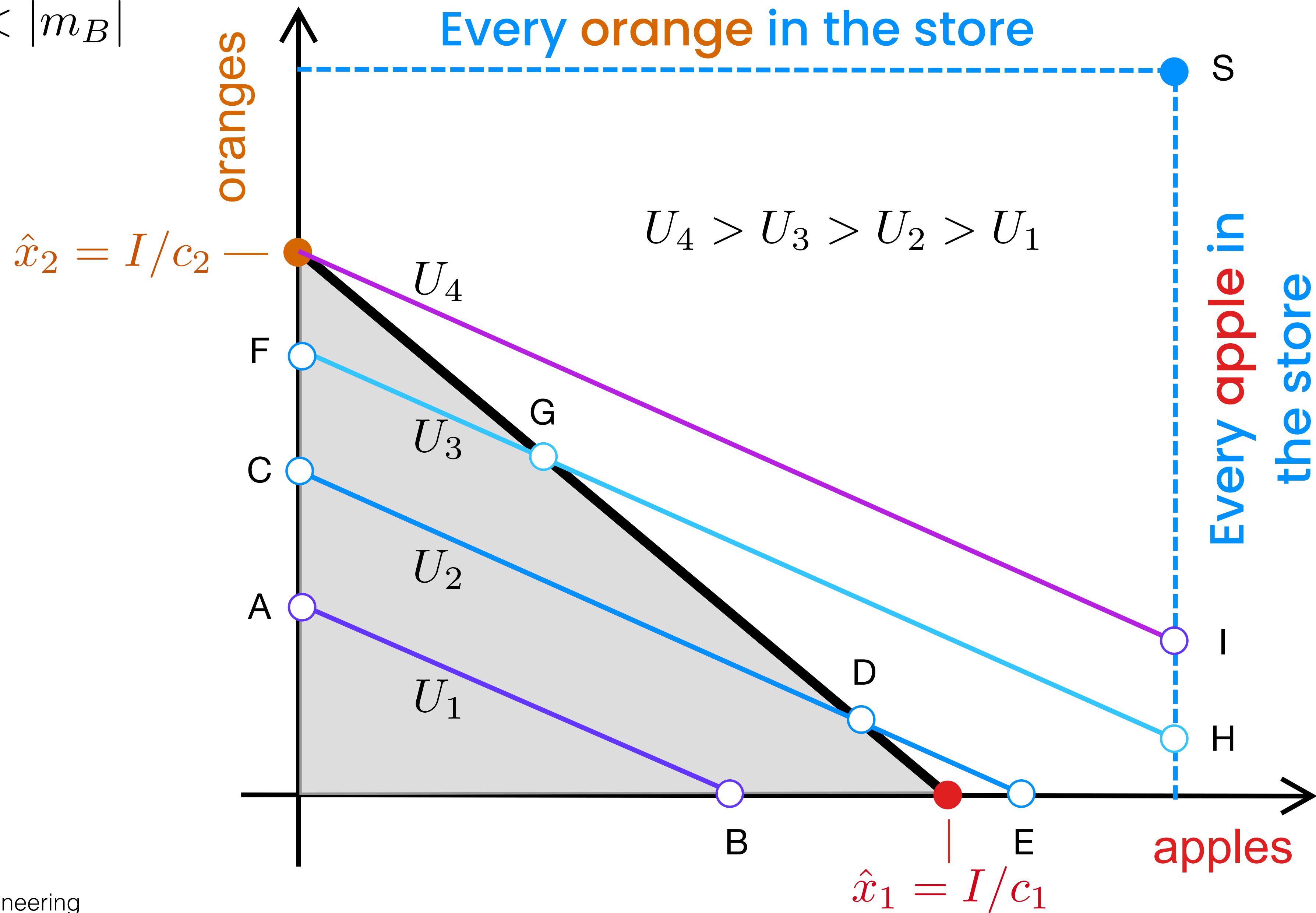
Utility: Linear (case A)

$|m_I| > |m_B|$



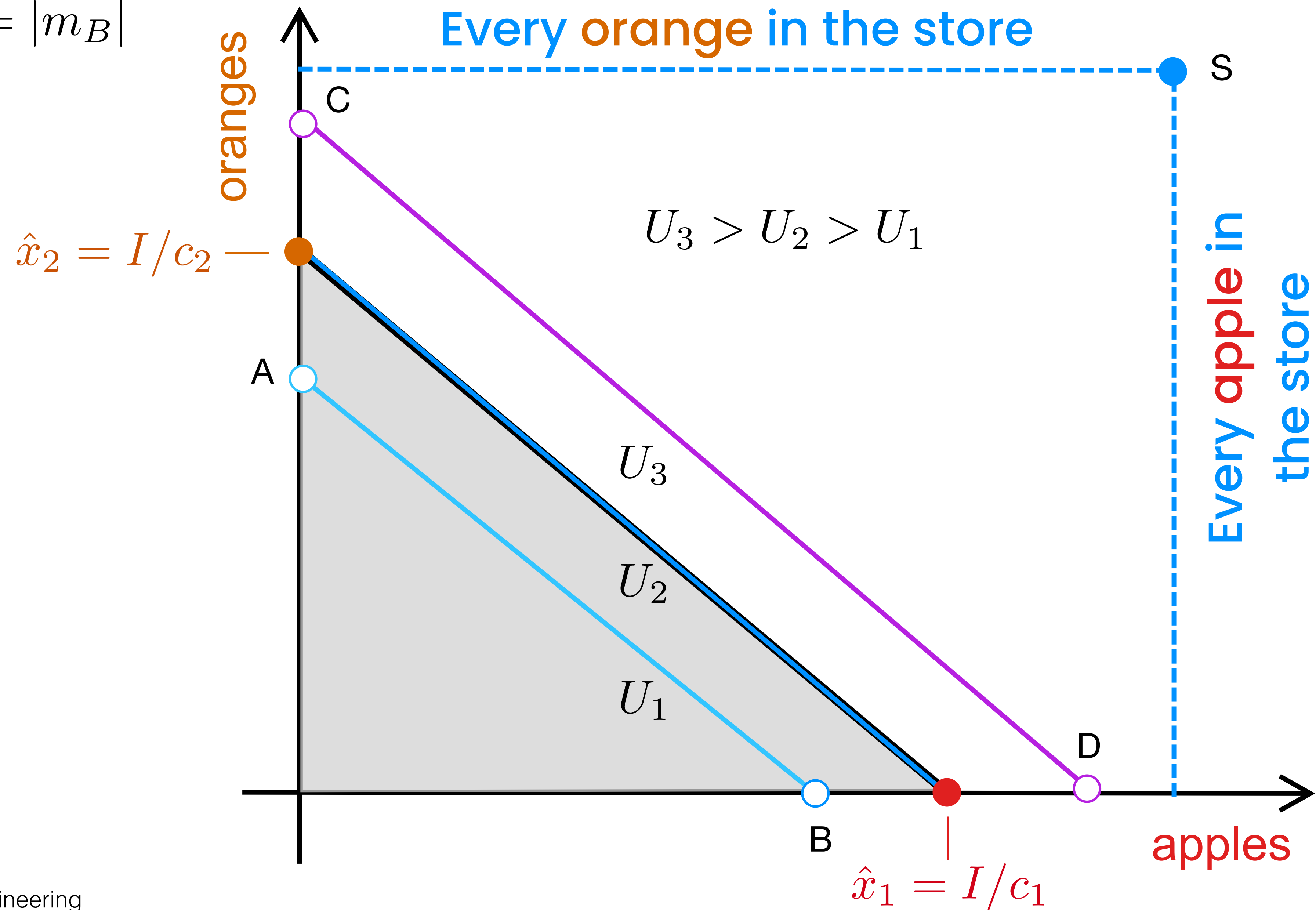
Utility: Linear (case B)

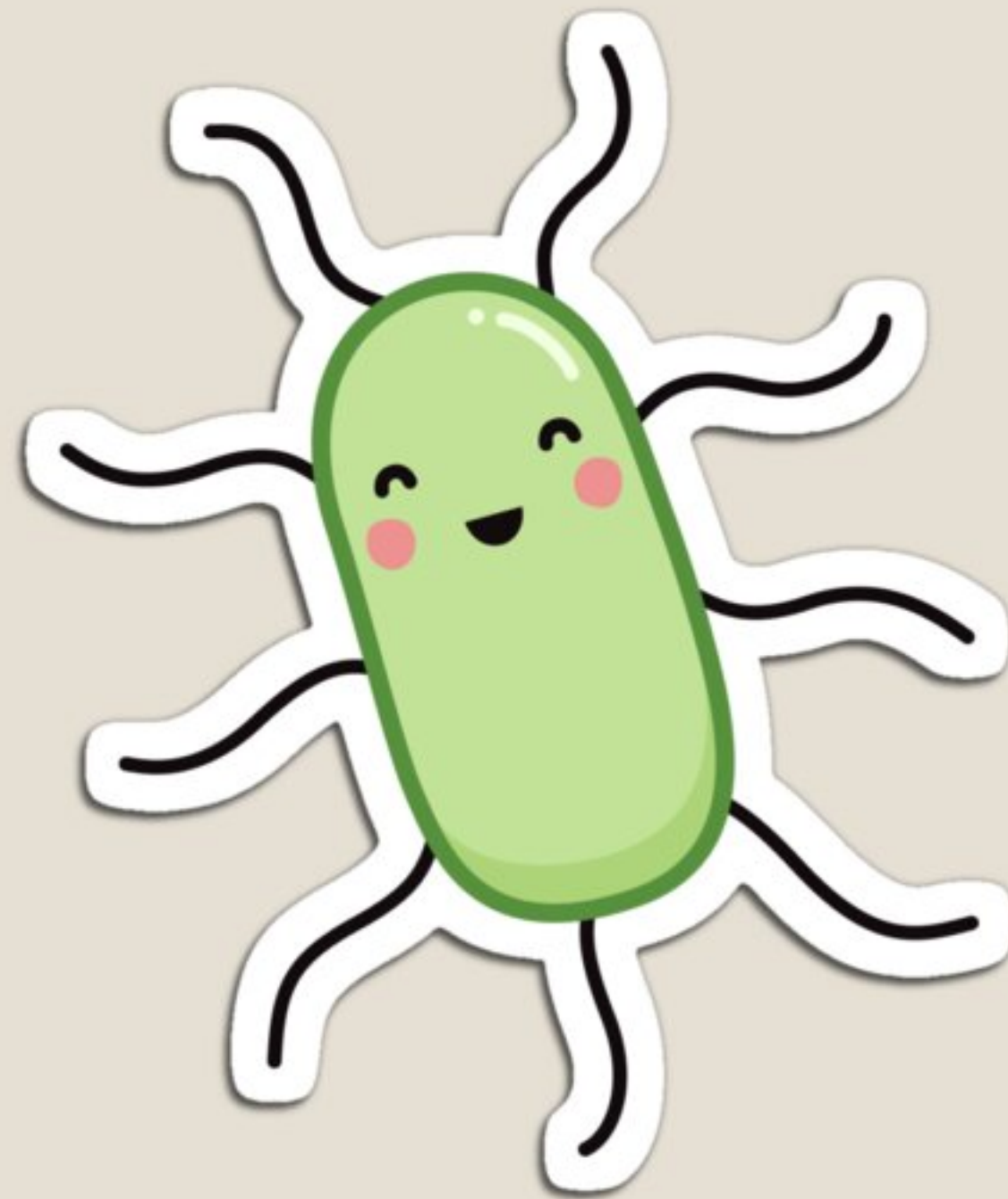
$|m_I| < |m_B|$



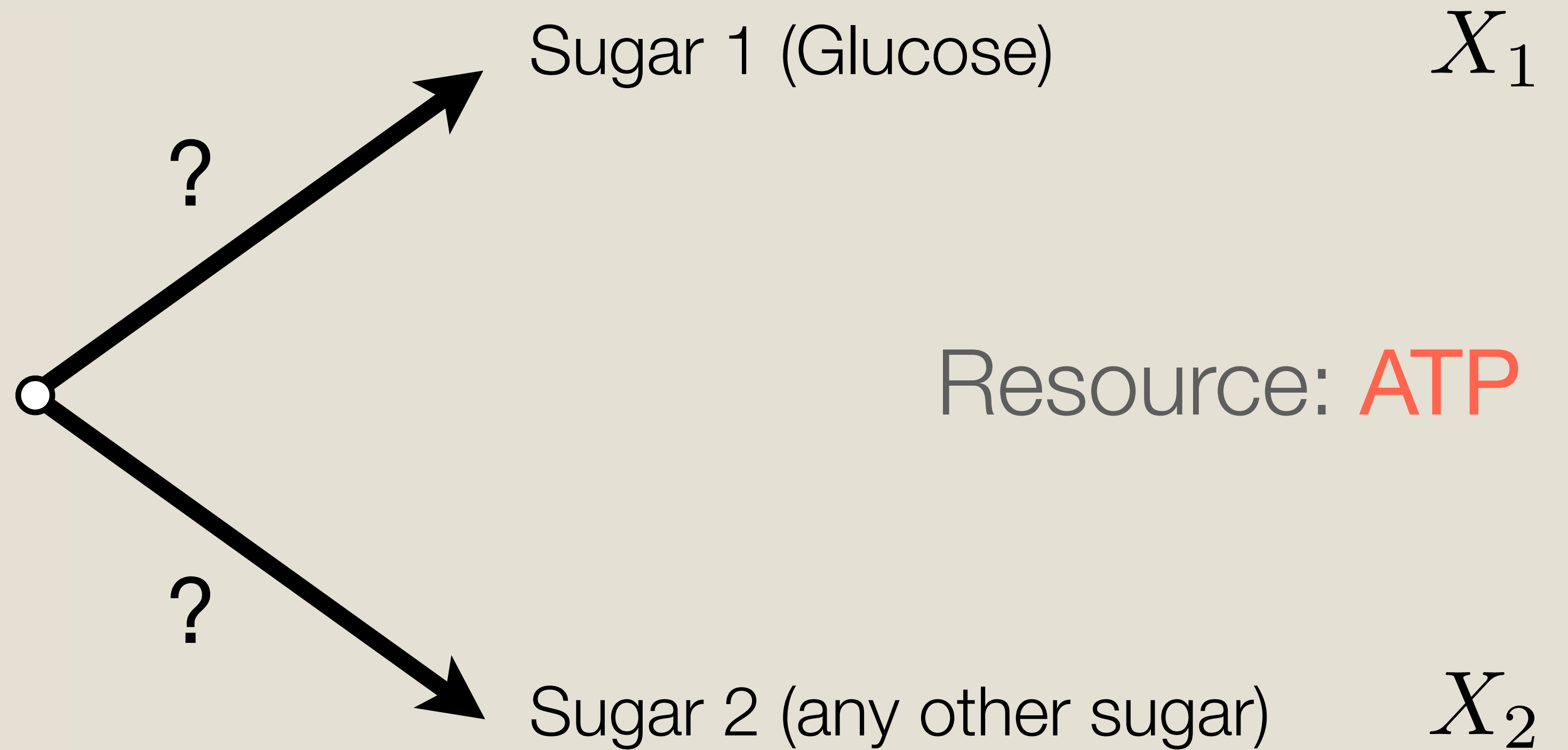
Utility: Linear (case C)

$$|m_I| = |m_B|$$





Escherichia coli



Today?